

1. Dataset Description

The dataset contains around 2000 grayscale medical images used for image classification. Each image represents one sample belonging to a specific class.

The images are not uniform in size. Some images contain black borders around the main region. The brightness and contrast levels vary between images. Because of these differences, the dataset needs preprocessing before training a deep learning model.

Main characteristics of the dataset:

- Around 2000 images
- Grayscale images
- Different image sizes
- Some images contain black background areas
- Variation in brightness and contrast

2. Data Preprocessing

Data preprocessing is done to clean and standardize the images before giving them to the model. The goal is to make all images consistent and improve their quality.

The following steps were applied:

1. CLAHE (Contrast Enhancement):

CLAHE is used to improve the contrast of grayscale images. It enhances important details without amplifying noise excessively.

2. Cropping:

Black borders around the main region are removed using thresholding and bounding box detection. This ensures the model focuses only on the relevant area.

3. Resizing:

All images are resized to 224×224 pixels to maintain uniform input size for the neural network.

4. Data Augmentation:

Mild brightness and small affine transformations are applied to increase variability and improve generalization.

5. Normalization:

Pixel values are scaled from 0–255 to 0–1 to stabilize model training.

3.YOLO Annotation Conversion

The YOLO part of the code converts bounding box information from a CSV file into YOLO format.

The CSV file contains image IDs and bounding box coordinates. For each image, the code reads its dimensions and calculates the bounding box center (x, y), width, and height.

These values are then normalized by dividing them by the image width and height, since YOLO requires all coordinates to be between 0 and 1.

Finally, the normalized values are saved in a .txt file with the same name as the image. Each image has a corresponding annotation file used for YOLO training.

4.Data Splitting

The dataset was divided into training and testing sets using a 70:30 ratio.

- 70% of the images are used for training.
- 30% of the images are used for testing.